Burundi: Debt Sustainability Analysis for Low-Income Countries Framework (LIC DSA)

1. This annex assesses the external and public debt dynamics of Burundi using the forward-looking framework for debt sustainability for low-income countries (LIC DSA), and draws some conclusions on the forward-looking sustainability of the country's debt.[1]

Baseline scenario

2. **The baseline scenario reflects continued growth of real GDP sustained by an increase in export volumes, and financing in the form of grants or highly concessional borrowing** (Box 5 in the main text). The medium-term outlook envisages achieving macroeconomic stability and solid economic growth, supported by a consolidation of the peace process, a deepening of structural reforms, and full normalization of relations with creditors. Exports are projected to drive growth over the long term based on the recovery of traditional export sectors and development of nontraditional exports. The current account deficit is projected to widen initially to above 10 percent of GDP in 2006, before narrowing to about 3.5 percent of GDP by 2024, while gross investment and national savings gradually rise to 13.7 percent and 10.1 percent of GDP, respectively, by 2024. Unless otherwise indicated, all debt ratios in this annex assume full delivery of HIPC assistance as proposed for the decision point and reflected in the HIPC DSA reported in the main text.

External debt sustainability assessment

3. **The results under the baseline scenario indicate that, even after full delivery of HIPC assistance, Burundi faces a considerable risk of debt distress.** The NPV of debt-to-exports ratio remains above the indicative threshold for countries with comparable policies and institutions for a considerable period of time before falling below it in 2020 (Box 1). Even though all other debt burden indicators are below their respective threshold during the projection period, the NPV of debt to exports ratio indicates that even after HIPC relief, and assuming future financing on highly concessional terms, that a debt distress situation would remain a concern over the next 15 years.

4. **At end-2004, Burundi's stock of public and publicly guaranteed (PPG) external debt was US$1,384 million, or US$895 in NPV terms.** Assuming the full delivery of HIPC assistance,[2] the NPV of debt would amount to US$76.3 million, corresponding to

[1] See "Debt Sustainability in Low-Income Countries: Proposal for an Operational Framework and Policy Implications" and "Debt Sustainability in Low-Income Countries: Further Considerations on an Operational Framework and Policy Implications." This DSA was prepared jointly by the staffs of the Fund and the Bank.

[2] The NPV of debt-to-exports ratio in the HIPC DSA methodology is 150 percent. Annex II Box 2, presents the methodological differences between the HIPC framework and the LIC framework in the calculation of the NPV debt-to-exports ratio, as well as the impact of such differences

13.8 percent of GDP, 155.6 percent of exports, and the ratio of debt service to exports would be 123.1 percent.[3] The debt sustainability indicators are projected to remain virtually unchanged or improve in 2005. Thereafter the profiles of the debt and debt service ratios over the long-term are as follows:

- ***The NPV of debt-to-exports indicator remains above the indicative threshold for much of the projection period, increasing over the medium term before gradually improving over the long-term*** (Table 1 and Figure 1). The NPV of debt-to-exports declines from 155.6 percent at end-2004 to 76.2 percent in 2024.

- ***The external debt service indicators suggest a more modest and manageable debt burden both in the short and medium terms.*** Public external debt service is projected to average about 9.6 percent of exports during the 2005-24 period and the projected trajectory shows an initial decline and then an increase through 2012, before decreasing to 6.1 percent in 2024 (Table 1 and Figure 1). Generally, the relatively low debt-service-to-exports indicators reflect: (i) a stable and relatively high grant element on the debt stock resulting in a relatively low debt service requirement, and (ii) high export growth during the period when debt service requirements increase as a result of the end of grace periods. The external debt service to fiscal revenues (including grants) follows a broadly similar trajectory and is well below the indicative threshold.

- ***The ratio of the NPV of debt to GDP is projected to*** **increase to 18.3 percent in 2009, before declining to 10 percent in 2024.**

5. **Alternative scenarios and stress tests indicate that lower growth, less concessional borrowing, and macroeconomic instability substantially increase the risk of debt distress**. Under the LIC framework an alternative scenario is shown in which the main parameters that determine the debt dynamics are assumed to remain at their historical averages. For Burundi, the period considered to calculate the historical averages is 2001-04, starting after the implementation of the Arusha Peace accord of August 2000. The historical scenario is intended to provide an indication about how optimistic the baseline is relative to the country's past performance during normal times. As shown in Figure 1, all debt burden indicators under the historical scenario present a significant deterioration compared to the baseline. In particular, the ratios of the NPV of debt to GDP and to exports exhibit an upward trend. The significant deterioration of debt and debt service indicators under the "historical scenario" underlines the critical importance of realizing the overall growth and export growth projections under the baseline.

6. **Stress tests signal the vulnerability of Burundi to adverse developments.** Burundi's debt sustainability indicators experience a considerable deterioration under stress

[3] Debt service in 2004 includes the repurchase of SDR 19.2 million previously drawn under the IMF's post-conflict emergency assistance policy.

tests assuming lower GDP and export growth or exchange rate depreciation (Table 2). For the NPV of debt-to-GDP ratio, there is a sizeable increase relative to the baseline when all the main parameters determining the debt dynamics are assumed to be one-half standard deviation below their respective historical averages during the period 2006-07. A significant increase in the external debt-to-exports and debt service-to-exports ratios is observed when exports are assumed to grow at the historical average minus one standard deviation.

Fiscal sustainability analysis

7. **The baseline macroeconomic scenario (Table 3 and Figure 2) is premised on increasing government revenues and financing the reconstruction effort and poverty reduction in the medium to long term primarily with grants.** This would allow the public sector borrowing requirement to fall over time, while keeping expenditures consistent with the government's poverty reduction strategy. In particular, it is assumed that revenues as a percentage of GDP rise from 18 percent of GDP in 2004 to 21 percent of GDP in 2024 and that borrowing decreases from 7.6 percent to 1 percent of GDP between 2005 and 2024.

8. **Given the relatively strong revenue effort, fiscal debt (external and domestic) burden indicators are projected to remain below their respective indicative thresholds.**[4] The NPV of debt-to-revenue ratio is projected to drop from 101.8 percent in 2004 to 49.6 percent in 2024, and the debt service-to-revenue ratio is projected to decline from 36.6 percent in 2004 to 12.3 percent in 2010 before dropping to 8.9 percent in 2024 (Table 3 and Figure 2).

9. **As in the external debt sustainability analysis, alternative scenarios and bound tests signal a significant increase in Burundi's risk of debt distress over the medium term.** The no reform scenario, which maintains the primary balance at the 2004 level, signals an increase in all debt indicators (Table 4). However, if the envisaged fiscal restraint is implemented, such a scenario would be a pessimistic benchmark. But, if there was no improvement in the fiscal primary balance after 2004, the NPV of debt-to-revenue indicator is projected to increase to 283 percent in 2024, a difference of 233 percentage points with respect to the baseline (Figure 2).

10. **Other stress tests highlight the negative impact of temporary shocks on debt burden indicators.** In particular, an exchange rate depreciation of 30 percent has the largest impact on debt stock indicators (Table 4 and Figure 2).

[4] In the LIC DSA methodology, the ratio of debt service to revenue is calculated with revenue including budgetary grants. Under the HIPC DSA framework, grants are excluded.

Box 1: Summary of Baseline Debt Sustainability Indicators 1/

	Indicative Threshold 2/	2005	2015	2024	Average 2005-24
NPV of debt to GDP	30	*13.9*	*136*	*9.9*	*14.1*
NPV of debt to exports	100	136.3	120.2	*76.2*	127.8
NPV of debt to revenue	200	*35.7*	*46.4*	*36.5*	*47.1*
Debt service to exports	15	35.9	*9.2*	*6.1*	*9.6*
Debt service to revenue	25	*9.4*	*3.5*	*2.9*	*3.4*

Note: Figures in italics are under the indicative thresholds.

1/ All debt indicators assume the full delivery of HIPC relief and refer to the NPV of public and publicly guaranteed external debt.
2/ Threshold over which countries with similar evaluations of policies and institutions would have at least a 25 percent chance of having a prolonged incident of debt distress in the coming year. Burundi lies within the bottom quintile of countries ranked by the World Bank's Country Policy and Institutional Assessment Index (CPIA).

Conclusion

11. **Burundi's NPV of external debt-to-exports ratio after enhanced HIPC assistance, is projected to remain above its indicative threshold over the period 2005-20. In addition, the DSA indicates that adverse shocks would negatively impact the debt sustainability indicators and would place Burundi in the high debt distress category for low-income countries.** The debt sustainability indicators for total (domestic and external debt) suggest a more robust position, reflecting a strong domestic revenue effort. However, as reflected in the external debt indicators, the foreign exchange constraint could be binding. Indeed, the external debt sustainability indicators suggest that particular attention be given to exports as the driver of overall growth.

12. **Sensitivity analysis reveals that if the pace of GDP (or export growth) falters, or the exchange rate depreciates, the debt burden could become unsustainable in the medium term.** The sensitivity analysis highlights the importance of sustained export-led growth and macroeconomic stability to decrease Burundi's risk of debt distress.

13. **The debt sustainability analysis indicates that in order for Burundi to lower its high debt stock ratios with respect to exports and to maintain the other debt indicators below their respective thresholds, particular emphasis needs to be placed on four aspects of future economic performance.** First, the authorities will need to implement a strong and sustained reform effort, especially to develop a diversified export base, to support robust growth in exports and GDP. Second, the authorities need to implement prudent fiscal and monetary policies to ensure fiscal discipline and macroeconomic stability. Third, together with prudent debt management, future financing needs to be mainly in the form of grants and highly concessional loans. Fourth, attention should be given to strengthening of policies and institutions, which would raise the indicative debt thresholds for the risk of debt distress.

Box 2. Impact of the Use of the LIC Framework Methodology on the End-2004 NPV of Debt-to-Exports Ratio

Using the new LIC framework methodology, Burundi's NPV of debt-to-exports ratio is estimated to have reached 155.6 percent at end-2004. This is 5.6 percentage points higher compared with the 150 percent obtained using the HIPC methodology, assuming full delivery of HIPC assistance. The increase is due to three factors:

(i) a 34 percentage points increase is due to the change in the exchange rates used to calculate the projected debt service streams (projected exchange rates from WEO; under the LIC framework versus end-2004 exchange rates under the HIPC methodology),

(ii) a 7.9 percentage points decrease is explained by the change in the discount rate used to calculate the NPV of debt (a single 5 percent discount rate under the LIC framework versus currency-specific discount rates under the HIPC methodology), and

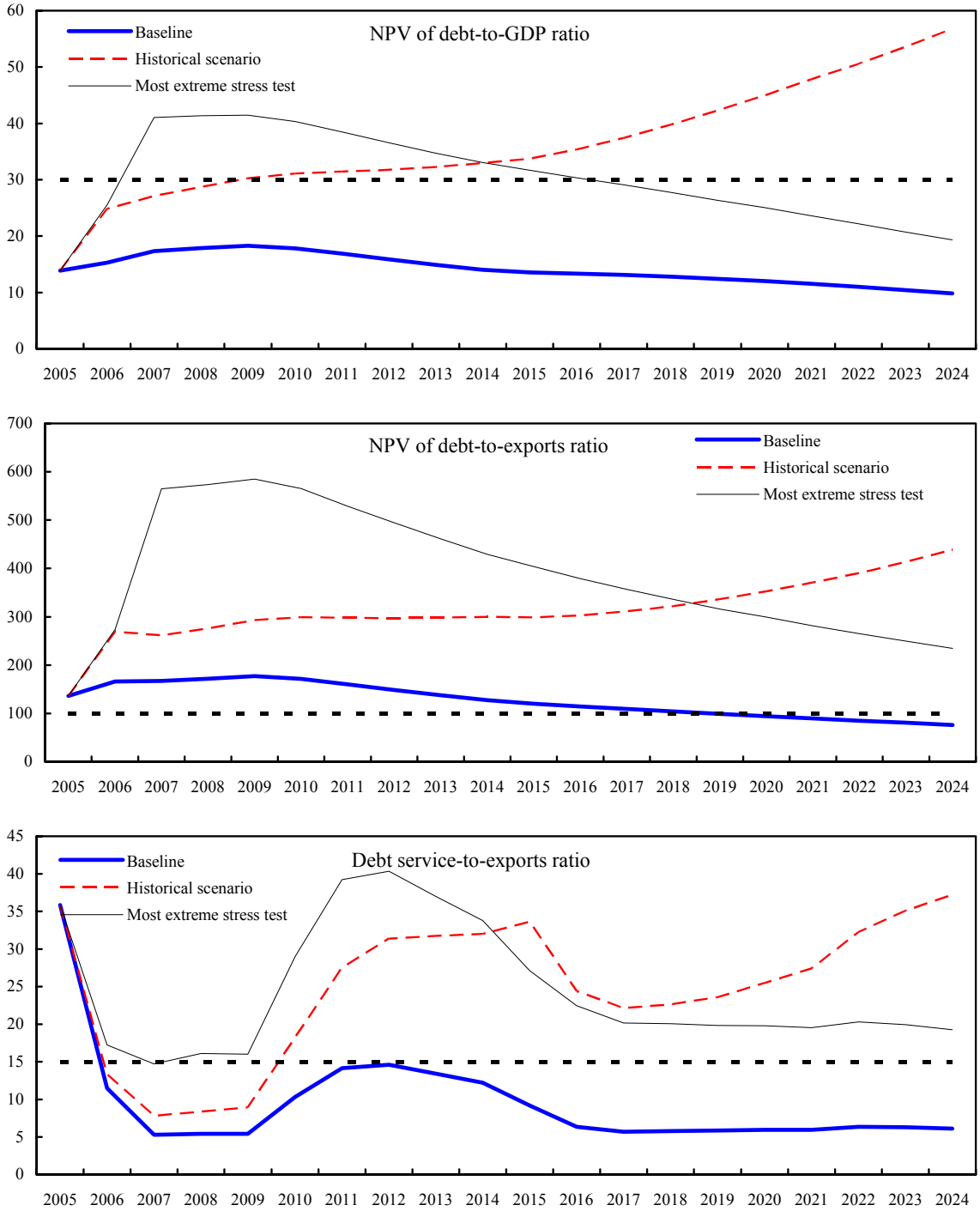
(iii) a 20.5 percentage points increase is due to the change in the denominator used to calculate the ratio (current exports under the LIC framework versus a three-year backward-looking average under the HIPC framework).

Burundi: Decomposition of the decrease in the NPV of debt-to-exports ratio at end-2004 1/
(in percentage points, unless otherwise indicated)

Factors explaining the change in the NPV of debt-to-exports ratio	Impact of methodological changes	
Total Change	**5.6**	
Of which: due to changes in:		
Exchange rates	34.0	
Discount rates	-7.9	
Exports of goods and services	-20.5	
Memorandum items:		
	Decision point	LIC framework
End-2004 NPV of debt-to-exports ratio	150.0	155.6
Exports of goods and services		
Current	58.9	58.9
Three year average	50.9	50.9

1/ Refers to public and publicly guarenteed external debt.

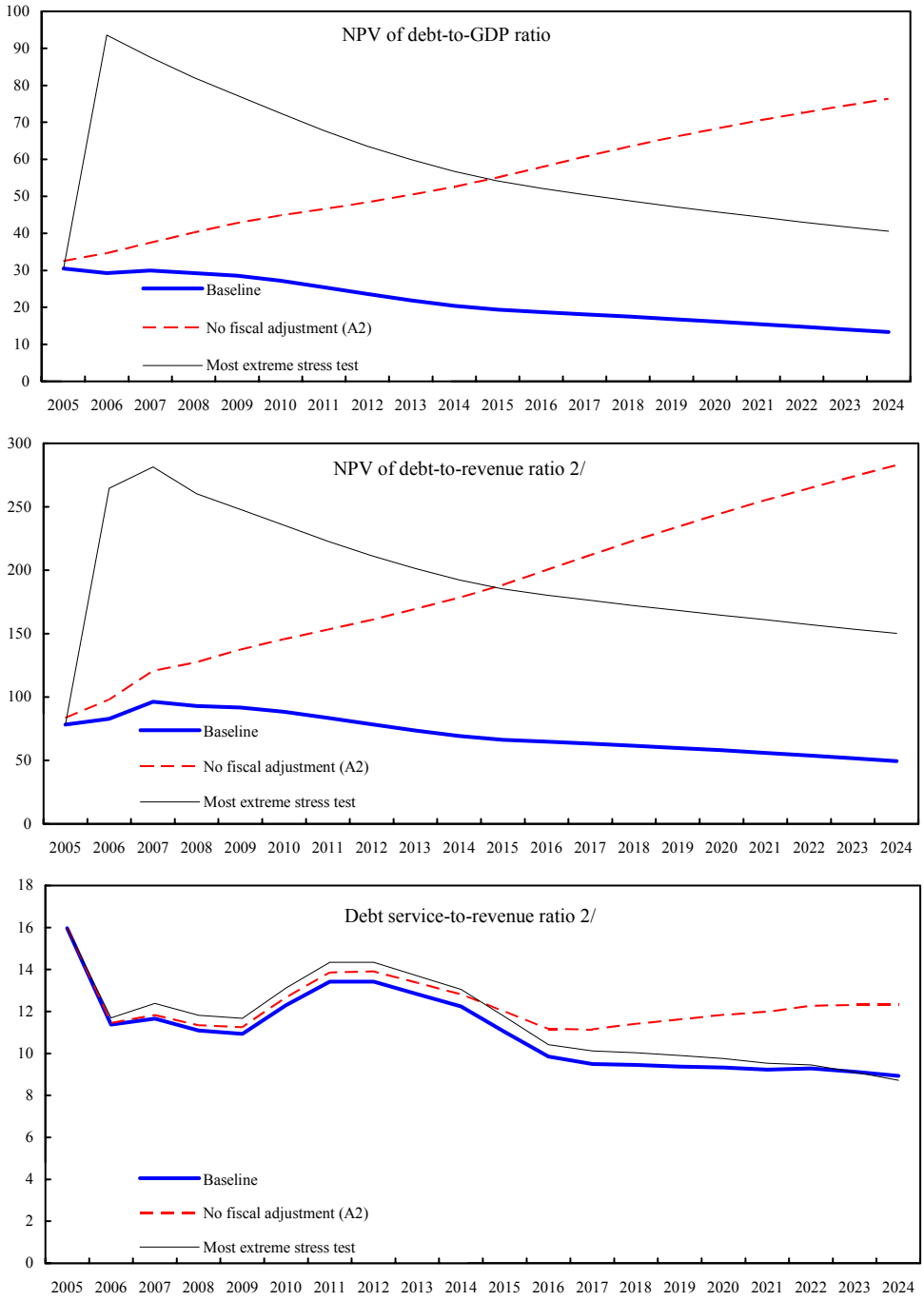
Figure 1. Burundi: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2005-2024 1/
(In percent)



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2015.

Figure 2. Burundi: Indicators of Public Debt Under Alternative Scenarios, 2005-24 1/

(In percent)



Source: Staff projections and simulations.
1/ Most extreme stress test is test that yields highest ratio in 2015.
2/ Revenue including grants.

Table 1. Burundi: External Debt Sustainability Framework, Baseline Scenario, 2004-24 1/
(In percent of GDP, unless otherwise indicated)

	Act. 2004	Historical Average 6/	Standard Deviation 6/	Est. 2005	Projections 2006	2007	2008	2009	2010	2005-10 Average	2015	2020	2024	2011-24 Average
External debt (nominal) 1/	208.4			178.2	34.8	37.1	37.2	37.0	35.7		27.2	22.4	18.0	
Of which: public and publicly guaranteed (PPG)	208.4			178.2	34.8	37.1	37.2	37.0	35.7		27.2	22.4	18.0	
Change in external debt	-18.7			-30.2	-143.5	2.3	0.1	-0.2	-1.3		-1.2	-0.9	-1.1	
Identified net debt-creating flows	-16.5			-1.4	1.7	7.4	5.9	5.8	5.4		3.5	2.4	1.9	
Noninterest current account deficit	5.9	4.4	2.4	6.1	10.2	9.9	8.5	8.4	7.9	8.5	5.6	4.2	3.4	5.0
Deficit in balance of goods and services	22.0			26.7	23.6	20.7	19.4	18.6	17.5		13.7	11.1	9.5	
Exports	8.9			10.2	9.2	10.4	10.4	10.3	10.4		11.3	12.8	13.0	
Imports	30.9			36.9	32.8	31.0	29.8	29.0	27.9		25.0	23.9	22.4	
Net current transfers (negative = inflow)	-17.2	-8.7	5.8	-21.8	-14.8	-12.2	-12.0	-11.2	-10.5	-13.8	-8.6	-7.2	-6.3	-8.0
Other current account flows (negative = net inflow)	1.0			1.3	1.5	1.4	1.2	1.0	0.9		0.5	0.3	0.2	
Net FDI (negative = inflow)	0.0	-0.2	0.5	-0.2	-1.1	-1.1	-1.1	-1.1	-1.1	-0.9	-0.9	-0.8	-0.8	-0.9
Endogenous debt dynamics 2/	-22.4			-7.3	-7.4	-1.4	-1.5	-1.5	-1.5		-1.1	-0.9	-0.8	
Contribution from nominal interest rate	1.3			1.3	0.5	0.3	0.3	0.2	0.2		0.2	0.1	0.1	
Contribution from real GDP growth	-9.8			-8.6	-7.9	-1.6	-1.7	-1.7	-1.7		-1.3	-1.1	-0.9	
Contribution from price and exchange rate changes	-13.8			…	…	…	…	…	…		…	…	…	
Residual (3-4) 3/	-2.2			-28.8	-145.2	-5.0	-5.8	-6.1	-6.6		-4.7	-3.4	-3.0	
Of which: exceptional financing	-16.0			-19.8	-134.2	-3.9	-3.8	-4.0	-4.2		-2.8	-1.8	-1.4	
NPV of external debt 4/	13.8			13.9	15.3	17.3	17.9	18.3	17.9		13.6	12.0	9.9	
In percent of exports	155.6			136.3	166.1	167.3	172.0	177.2	171.5		120.2	94.4	76.2	
NPV of PPG external debt	13.8			13.9	15.3	17.3	17.9	18.3	17.9		13.6	12.0	9.9	
In percent of exports	155.6			136.3	166.1	167.3	172.0	177.2	171.5		120.2	94.4	76.2	
Debt service-to-exports ratio (in percent)	123.1			35.9	11.5	5.3	5.4	5.4	10.3		9.2	6.0	6.1	
PPG debt service-to-exports ratio (in percent)	123.1			35.9	11.5	5.3	5.4	5.4	10.3		9.2	6.0	6.1	
Total gross financing need (billions of U.S. dollars)	111.7			1516.5	1373.3	89.9	83.0	88.9	96.2		100.3	102.3	111.7	
Noninterest current account deficit that stabilizes debt ratio	24.6			36.3	153.7	7.5	8.4	8.7	9.2		6.8	5.2	4.5	
Key macroeconomic assumptions														
Real GDP growth (in percent)	4.8	1.7	2.7	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
GDP deflator in U.S. dollar terms (change in percent)	6.5	-4.5	8.8	14.8	7.3	2.4	2.4	2.5	2.8	5.4	2.2	2.0	2.0	2.2
Effective interest rate (percent) 5/	0.6	1.0	0.2	0.7	0.3	0.8	0.8	0.6	0.7	0.7	0.7	0.7	0.7	0.7
Growth of exports of goods and services (U.S. dollar terms, in percent)	7.0	7.0	39.3	38.5	1.8	20.9	8.1	6.7	8.8	14.1	10.1	8.6	7.1	9.0
Growth of imports of goods and services (U.S. dollar terms, in percent)	19.9	6.4	16.0	43.8	0.2	1.7	3.2	4.8	4.0	9.6	5.9	6.7	5.6	5.6
Grant element of new public sector borrowing (in percent)	...	...	...	45.6	41.0	48.5	54.6	54.4	54.2	49.7	53.9	54.0	53.7	54.0
Memorandym item:														
Nominal GDP (billions of U.S. dollars)	664.1			800.4	901.6	969.3	1043.0	1122.9	1211.8		1751.4	2466.2	3239.8	

Source: Staff projections and simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are derived over the past 4 years (from 2001), subject to data availability.

Table 2. Burundi: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-24
(In percent)

	Est.	Projections						
	2005	2006	2007	2008	2009	2010	2015	2024
NPV of debt-to-GDP ratio								
Baseline	14	15	17	18	18	18	14	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2005-24 1/	14	25	27	29	30	31	34	57
A2. New public sector loans on less favorable terms in 2005-24 2/	14	16	19	21	22	22	19	16
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	14	16	20	20	21	20	15	11
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	14	17	22	22	22	22	17	11
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2006-07	14	19	25	26	27	26	20	14
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2006-07 4/	14	22	29	29	29	28	22	13
B5. Combination of B1-B4 using one-half standard deviation shocks	14	26	41	41	41	40	32	19
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	14	21	24	25	25	25	19	14
NPV of debt-to-exports ratio								
Baseline	136	166	167	172	177	172	120	76
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2005-24 1/	136	269	262	276	293	299	299	438
A2. New public sector loans on less favorable terms in 2005-24 2/	136	177	187	199	209	208	172	127
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	136	166	167	172	177	172	120	76
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	136	274	565	574	585	565	404	234
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2006-07	136	166	167	172	177	172	120	76
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2006-07 4/	136	235	276	277	279	269	195	104
B5. Combination of B1-B4 using one-half standard deviation shocks	136	262	440	442	447	431	312	166
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	136	166	167	172	177	172	120	76
Debt service ratio								
Baseline	36	11	5	5	5	10	9	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2005-24 1/	36	13	8	8	9	18	34	37
A2. New public sector loans on less favorable terms in 2005-24 2/	36	11	6	7	7	9	10	8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	36	11	5	5	5	10	9	6
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	36	17	15	16	16	29	27	19
B3. U.S. dollar GDP deflator at historical average minus one standard deviation in 2006-07	36	11	5	5	5	10	9	6
B4. Net non-debt-creating flows at historical average minus one standard deviation in 2006-07 4/	36	11	6	7	7	12	13	9
B5. Combination of B1-B4 using one-half standard deviation shocks	36	13	10	11	11	19	20	14
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	36	11	5	5	5	10	9	6
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	53	53	53	53	53	53	53	53

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Burundi: Public Sector Debt Sustainability Framework, Baseline Scenario, 2004-24
(In percent of GDP, unless otherwise indicated)

	Act. 2004	Historical Average 5/	Standard Deviation 5/	Est. 2005	Projections 2006	2007	2008	2009	2010	2005-10 Average	2015	2020	2024	2011-24 Average
Public sector debt 1/	232.3			194.2	49.5	50.4	49.3	47.8	45.5		33.4	26.9	21.7	
Of which: foreign-currency denominated	210.0			177.3	35.3	37.6	37.7	37.4	36.1		27.5	22.7	18.1	
Change in public sector debt	-15.1			-38.2	-144.7	0.9	-1.1	-1.4	-2.3		-1.8	-1.2	-1.3	
Identified debt-creating flows	-38.0			-61.8	-155.1	-5.2	-5.7	-5.8	-6.2		-6.4	-6.7	-7.0	
Primary deficit	1.0	2.2	2.9	-3.6	-4.9	-3.8	-5.0	-5.0	-5.3	-4.6	-6.3	-6.8	-7.2	-6.5
Revenue and grants	35.5			38.9	35.4	31.1	31.5	31.2	30.8		29.3	27.9	27.0	
Of which: grants	15.4			20.7	16.8	12.3	12.5	12.0	11.4		9.0	7.3	6.2	
Primary (noninterest) expenditure	36.5			35.3	30.5	27.3	26.5	26.1	25.5		23.0	21.1	19.8	
Automatic debt dynamics	-23.0			-38.1	-15.1	-0.6	-0.7	-0.8	-0.8		-0.2	0.1	0.2	
Contribution from interest rate/growth differential	-13.8			-14.3	-10.4	-0.4	-0.5	-0.5	-0.5		-0.1	0.1	0.2	
Of which: contribution from average real interest rate	-2.4			-3.3	-1.1	2.0	1.9	1.8	1.8		1.6	1.4	1.3	
Of which: contribution from real GDP growth	-11.4			-11.0	-9.3	-2.4	-2.4	-2.4	-2.3		-1.7	-1.3	-1.1	
Contribution from real exchange rate depreciation	-9.2			-23.8	-4.7	-0.2	-0.2	-0.3	-0.3		...	...	...	
Other identified debt-creating flows	-16.0			-20.1	-135.2	-0.8	0.0	0.0	0.0		0.0	0.0	0.0	
Privatization receipts (negative)	0.0			-0.3	-1.0	-0.8	0.0	0.0	0.0		0.0	0.0	0.0	
Recognition of implicit or contingent liabilities	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	
Debt relief (HIPC and other)	-16.0			-19.8	-134.2	0.0	0.0	0.0	0.0		0.0	0.0	0.0	
Other (specify, e.g., bank recapitalization)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	0.0	
Residual, including asset changes	22.9			23.6	10.5	6.1	4.6	4.4	3.9		4.7	5.5	5.7	
NPV of public sector debt	36.1			30.7	29.5	30.1	29.4	28.7	27.3		19.5	16.3	13.4	
Of which: foreign-currency denominated	13.8			13.9	15.3	17.3	17.9	18.3	17.9		13.6	12.0	9.9	
external	13.8			13.9	15.3	17.3	17.9	18.3	17.9		13.6	12.0	9.9	
NPV of contingent liabilities (not included in public sector debt)	0.0			6.8	6.0	5.4	4.9	4.4	4.0		2.5	1.6	1.1	
Gross financing need 2/	9.6			2.4	0.5	0.3	0.3	0.3	0.8		0.9	0.6	0.7	
NPV of public sector debt-to-revenue ratio (in percent) 3/	101.8			78.9	83.4	96.8	93.4	92.2	88.8		66.7	58.4	49.6	
Of which: external	38.9			35.7	43.3	55.7	56.8	58.7	58.0		46.4	43.2	36.5	
Debt service-to-revenue ratio (in percent) 3/ 4/	36.6			16.0	11.4	11.7	11.1	10.9	12.3		11.0	9.3	8.9	
Primary deficit that stabilizes the debt-to-GDP ratio	16.1			34.6	139.8	-4.7	-3.9	-3.6	-3.0		-4.5	-5.6	-5.9	
Key macroeconomic and fiscal assumptions														
Real GDP growth (in percent)	4.8	1.7	2.7	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0
Average nominal interest rate on forex debt (in percent)	0.6	1.1	0.2	0.7	0.3	0.8	0.8	0.6	0.7	0.7	0.7	0.7	0.7	0.7
Average real interest rate on domestic currency debt (in percent)	4.3	-1.1	8.9	-1.2	10.7	17.6	19.0	21.0	22.5	14.9	31.2	39.4	42.4	34.7
Real exchange rate depreciation (in percent, indicates depreciation +)	-4.3	4.1	14.2	-12.1	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	8.3	12.0	8.3	15.3	8.4	5.5	5.4	5.2	5.0	7.5	4.3	4.0	4.0	4.2
Growth of real primary spending (deflated by GDP deflator, in percent)	23.2	7.2	18.2	1.6	-9.3	-5.9	2.0	3.4	2.3	-1.0	3.5	3.7	3.5	3.1
Grant element of new external borrowing (in percent)	...	0.0	0.0	45.6	41.0	48.5	54.6	54.4	54.2	49.7	53.9	54.0	53.7	...

Sources: Country authorities; and staff estimates and projections.

1/ Burundi's public debt is defined as external public and publicly guaranteed debt and domestic central government debt, gross basis.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally over the past four years (from 2001), subject to data availability.

Table 4. Burundi: Sensitivity Analysis for Key Indicators of Public Debt, 2005-24

	Est.	Projections							
	2005	2006	2007	2008	2009	2010	2015	2020	2024
NPV of Debt-to-GDP Ratio									
Baseline	31	29	30	29	29	27	20	16	13
A. Alternative scenarios									
A1. Real GDP growth and primary balance are at historical averages	31	34	37	41	45	48	67	93	115
A2. Primary balance is unchanged from 2004	33	35	38	40	43	45	55	68	76
A3. Permanently lower GDP growth 1/	31	30	31	32	32	32	32	37	41
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2006-07	31	32	36	37	38	38	39	43	45
B2. Primary balance is at historical average minus one standard deviations in 2006-07	31	34	37	37	37	36	33	32	31
B3. Combination of B1-B2 using one half standard deviation shocks	31	34	39	39	39	38	33	33	32
B4. One-time 30 percent real depreciation in 2006	31	94	88	82	77	73	54	46	41
B5. 10 percent of GDP increase in other debt-creating flows in 2006	31	35	35	35	35	35	31	31	30
NPV of Debt-to-Revenue Ratio 2/									
Baseline	79	83	97	93	92	89	67	58	50
A. Alternative scenarios									
A1. Real GDP growth and primary balance are at historical averages	79	94	117	126	138	149	210	300	375
A2. Primary balance is unchanged from 2004	84	99	121	128	138	146	189	245	283
A3. Permanently lower GDP growth 1/	79	86	101	101	103	104	109	131	148
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2006-07	79	89	110	112	116	119	129	152	163
B2. Primary balance is at historical average minus one standard deviations in 2006-07	79	95	119	118	119	118	112	116	114
B3. Combination of B1-B2 using one half standard deviation shocks	79	95	122	119	120	119	111	115	116
B4. One-time 30 percent real depreciation in 2006	79	265	282	261	249	236	186	165	150
B5. 10 percent of GDP increase in other debt-creating flows in 2006	79	98	114	112	114	113	107	112	111
Debt Service-to-Revenue Ratio 2/									
Baseline	16	11	12	11	11	12	11	9	9
A. Alternative scenarios									
A1. Real GDP growth and primary balance are at historical averages	16	12	12	12	12	14	14	15	16
A2. Primary balance is unchanged from 2004	16	11	12	11	11	13	12	12	12
A3. Permanently lower GDP growth 1/	16	11	12	11	11	12	11	9	8
B. Bound tests									
B1. Real GDP growth is at historical average minus one standard deviations in 2006-07	16	12	12	12	12	13	12	10	9
B2. Primary balance is at historical average minus one standard deviations in 2006-07	16	11	12	11	11	12	11	9	7
B3. Combination of B1-B2 using one half standard deviation shocks	16	12	12	12	12	13	11	9	7
B4. One-time 30 percent real depreciation in 2006	16	12	12	11	11	13	11	8	7
B5. 10 percent of GDP increase in other debt-creating flows in 2006	16	11	12	11	11	12	11	9	7

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.